



TIR-Technical Information Release Process

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

Technical Information Release (TIR) is a Caterpillar Dealer produced document used to communicate product problems, interim resolutions, dealer repair/setup procedures with appropriate groups within the dealership. TIR's are an effective and efficient path to inform multiple areas of the dealer, customers, and 3rd parties (where applicable). This best practice will outline and explain TIR structure, data collection and distribution.

2.0 Best Practice Description

This Best Practice provides guidelines on how a TIR should be structured to effectively communicate information, display illustrations, capture data, document accessibility and distribution among the Dealership.

3.0 Implementation Steps

TIR Document Structure:

- TIR Header should include Dealer name and confidential level
- Media number required to track and identify each TIR – the use of the publication date may be an option (example: TIR2021-11-15)
- Publication Date – Date of TIR Publication
- Date Updated – Date of latest revision
- Distribution shows the confidential level of the TIR such as “Dealer only”, “Customers”, “3rd Party”
- Priority shows severity of the TIR such as “Information Only”, “Take Necessary Precautions”, “Immediate Action”

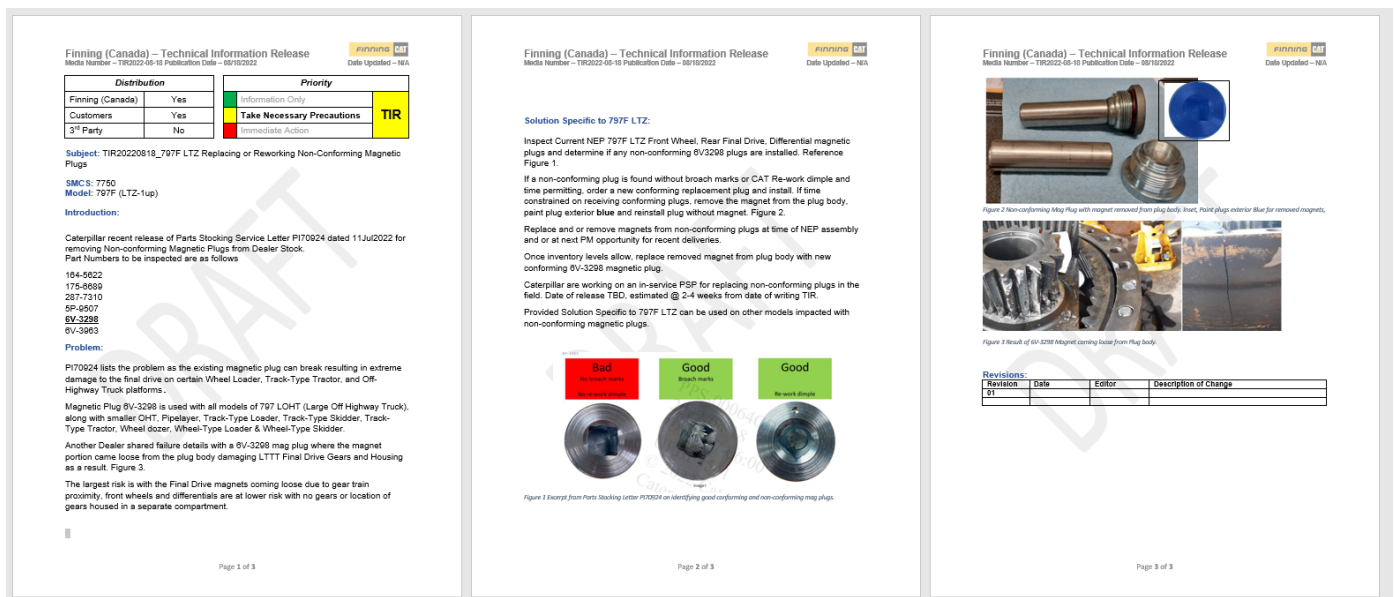


Figure 1 - Technical Information Release Example

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Subject – Description of the Technical Information Release

SMCS – SMCS Codes applicable to the TIR

Models – Affected Prefix / Serial numbers

Introduction – Elaborate on the subject, provide brief background information

Problem – Outline the issue, include potential contingent damage

Solution – Outline interim/permanent solutions, include action items if required

Data Capture Table – Failure information requested and communicated back to the technical department (If Applicable)

Distribution		Priority		Colour coded confidential level for quick reference TIR
Finning (Canada)	Yes	Information Only		
Customers	Yes	Take Necessary Precautions		
3 rd Party	No	Immediate Action		

TIR Introduction:

This section is intended to provide background information on the technical information release. Details as to why this is required and how it will be accomplished.

TIR Problem:

This section provides details on the problem outlined by the TIR. Clear and descriptive information is required to communicate historic, current and potential issues this problem may cause. If possible, include (pictures, diagrams, drawings etc) to ensure the problem is fully understood by the intended audience.

TIR Solution.

Parts/Tools Required:

Table of all parts and tooling required to complete the Technical Information Release. This will ensure parts and tooling can be ordered ahead of time and planned accordingly before the work gets done. It will also be used for future reference when the particular TIR requires repairs or maintenance related tasks.

Reference	Part Number	Description	Quantity
A	328002044	Top Bracket	1
B	328002045	Bottom Bracket	1
C	Bolt (1"-8")	7X0415	4
D	Stover Nut	6K-3632	4
E	FTA-TSB-4HD-50	Rubber Gasket (7ft)	2

Figure 2 - Example TIR Parts List

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Installation Procedure:

Step by step instructions to ensure the Technical Information Release is completed efficiently and in a repeatable manner on each applicable machine. If needed, provide pictures, drawings and schematics to clarify the procedure.

***** List any reference documents related to the Technical Information Release**

Safety Section:

This section is intended to provide any precautions that should be taken during the inspection, repair or other procedure outlined in the Technical Information Release.

Example:

Follow site-based safety recommendations.

Review referenced literature and follow safety recommendations.

Before you perform any repairs or before you perform any maintenance, read all safety information. Understand all safety information before you perform any repairs or before you perform any maintenance.



WARNING

Do not attempt to disassemble or assemble this machine until you read and understand the disassembly and assembly instructions. Improper disassembly and assembly procedures could result in injury or death.

TIR Conclusion:

Quick summary of the problem, solution, and general recommendations to implement the TIR.

TIR Revisions:

Enter details of any revision made to this TIR Rev (01,02,03 etc)

Revision	Date	Editor	Description of Change
01			

TIR Risk Assessment:

Utilize a risk matrix to assess any associated risks involved with the Technical Information Release. Risks during the installation process, machine operation and maintenance requirements for the projected life of the equipment.

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RISK ASSESSMENT

Step 1 - Determine Probability

A= Practically impossible
B= Not likely to occur
C= Could occur or I've heard of it happening
D= Is known to occur or "it has happened"
E= Common or frequent occurrence

Step 2 - Determine Consequences (highest of the two)

People
1= First aid injury
2= Medical treatment injury
3= Disabling short term lost time injury
4= Serious lost time injury or illness
5= Fatality or permanent disability

Plant, Property, Productivity or Environmental Consequences
1= Under \$5K damage and minimal productivity disruption, no environmental damage
2= \$5-\$50K damage and/or slight productivity disruption, minor environmental damage
3= \$50-\$100K damage and/or productivity disruption, reversible environmental damage
4= \$100-\$500K damage and/or contingency plan required, serious environmental damage
5= More than \$500K damage and/or large reorganization of premises, major environmental damage

Step 3 – Calculate the Risk					
	A	B	C	D	E
1	Low	Low	Low	Low	Low
2	Low	Low	Low	Moderate	Moderate
3	Low	Low	Moderate	Moderate	High
4	Low	Moderate	Moderate	High	High
5	Low	Moderate	High	High	High

Note: Risks ranked *Moderate* or *High* require supervisor involvement before work is started

Figure 3 - Example of Risk Assessment Matrix

Could it affect any of the following:

- Safety Devices – Handrails/Towing/Lifting/Tie-offs/Lock Out
- ROPS/FOPS
- Steering
- Brakes
- Egress

Perform Peer Review to assess all potential risks associated with the TIR. With any Technical Information Release, a validation period may be necessary to monitor how machine performance is affected.

TIR Capture Form:

If applicable, provide a template capture form to gather data from the field. Provide instructions on how to fill out the capture form in the TIR and how to communicate the information back to the appropriate contacts.

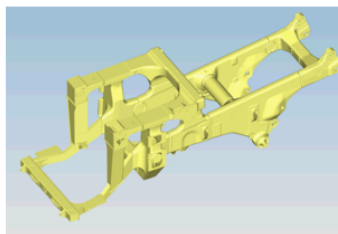


Figure 4 LH Frame View

Date of Inspection	
Work Order Number	
Unit Number	
Model	
Serial #	
SMU	
Crack Location	
Crack Length	
Casting Part #	
Casting Numerical Code	
Foundry Markings	
Picture of Crack	
Crack Location	
Crack Length	
Casting Part #	
Casting Numerical Code	
Foundry Markings	
Picture of Crack	
Crack Location	
Crack Length	
Casting Part #	
Casting Numerical Code	
Foundry Markings	
Picture of Crack	

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TIR Illustrations:

The use of relevant illustrations helps communicate information efficiently. Ensure to use arrows and descriptions to point out the correct part of the illustration to avoid misinterpretation.

- Refer to the operation manual for the device that you are using for advice on how to achieve quality photos and videos.
- Allow for ample lighting to reduce the number of photos that will be discarded due to blurriness.
 - Try to use natural light whenever possible.
 - Good lighting will also reduce or prevent shadows from developing on the components you are photographing.
- Capturing Photo, Video, Or Audio Evidence of Machine Issues
- Review the photos before forwarding them off to ensure that the photos show the problem clearly.

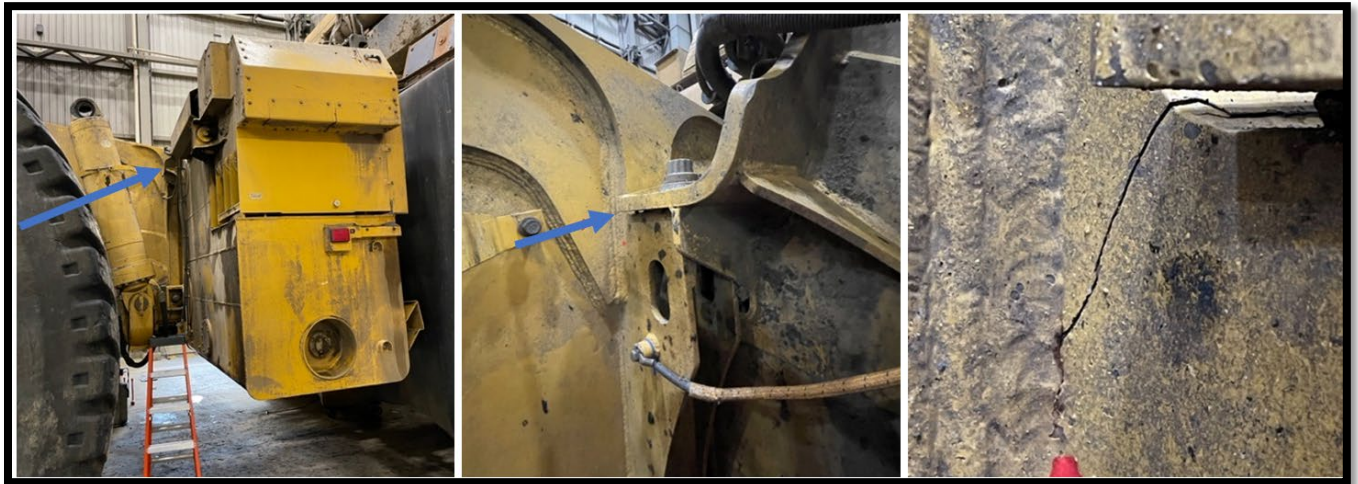


Figure 4 - Example Photo Showing Exact location with multiple pictures and arrows

TIR Distribution / Communication:

Technical Information Release has a wide range audience, including service, technical and depending on the issue, parts department. It is recommended that a TIR email distribution list is created to ensure consistent communication. Other platforms such as MS Teams, SharePoint and other shared sites are recommended for extended end-user reach.

Draft version of the TIR is sent to the Technical Team for peer review, discuss any background information, DPPM and CPI Status. Review any special procedures or tooling required to complete the work.

All current TIR's are reviewed by the Technical Team on a quarterly basis to ensure the latest information is communicated and outlines the best solution currently available.

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Communication Path (Examples)

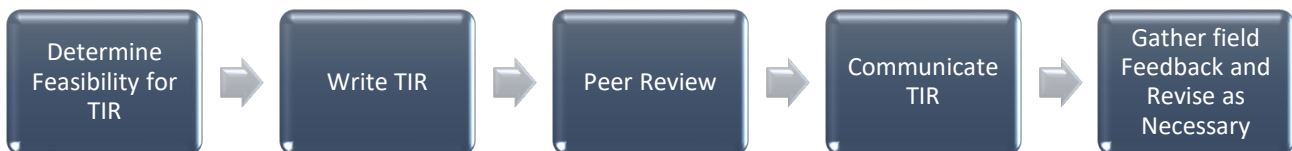
- Email - Send email to appropriate contacts
- MS Teams – Create a Library of TIR’s within a dealer MS Teams Site
- SharePoint – Create a Library of TIR’s on a dealer SharePoint Site.
- Technical Meeting – Share and discuss newly published TIR’s in customer technical meetings.

4.0 Benefits

- Ensures crucial information is communicated effectively and efficiently
- Easy and quick to publish
- Data is returned consistently to the technical team
- Early warning process for potential emerging issues

5.0 Resources Required

Technical Subject Matter Expert (Dealer)



6.0 Supporting Attachments / References

(TIR) Technical Information Release Template - Attachment

7.0 Related Best Practices

None

8.0 Acknowledgements

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